Mathematical Models

21. Motion in a Force Field A mathematical model for the position $x(t)$ of a body moving rectilinearly on the x-axis in an inverse-square force field is given by

$$\frac{d^2x}{dt^2} = -\frac{k^2}{x^2}.$$

Suppose that at $t \quad 0$ the body starts from rest from the position $x \quad x_0$, $x_0 > 0$. Show that the velocity of the body at time t is given by $v^2 \quad 2k^2(1/x - 1/x_0)$. Use the last expression and a CAS to carry out the integration to express time t in terms of x.

22. A mathematical model for the position $x(t)$ of a moving object is

$$\frac{d^2x}{dt^2} + \sin x \quad 0.$$

Use a numerical solver to graphically investigate the solutions of the equation subject to $x(0) \quad 0, x'(0) \quad x_1, x_1 \geq 0$. Discuss the motion of the object for $t \geq 0$ and for various choices of x_1. Investigate the equation

$$\frac{d^2x}{dt^2} + \frac{dx}{dt} + \sin x \quad 0$$

in the same manner. Give a possible physical interpretation of the dx/dt term.

3.8 Linear Models: Initial-Value Problems

Introduction In this section we are going to consider several linear dynamical systems in which each mathematical model is a linear second-order differential equation with constant coefficients along with initial conditions specified at time t_0:

$$a_2\frac{d^2y}{dt^2} + a_1\frac{dy}{dt} + a_0y \quad g(t), \quad y(t_0) \quad y_0, \quad y'(t_0) \quad y_1.$$

Recall, the function g is the **input**, **driving**, or **forcing function** of the system. The **output** or **response** of the system is a function $y(t)$ defined on an I interval containing t_0 that satisfies both the differential equation and the initial conditions on the interval I.

3.8.1 Spring/Mass Systems: Free Undamped Motion

Hooke's Law Suppose a flexible spring is suspended vertically from a rigid support and then a mass m is attached to its free end. The amount of stretch, or elongation, of the spring will, of course, depend on the mass; masses with different weights stretch the spring by differing amounts. By Hooke's law, the spring itself exerts a restoring force F opposite to the direction of elongation and proportional to the amount of elongation s. Simply stated, $F \quad ks$, where k is a constant of proportionality called the **spring constant**. The spring is essentially characterized by the number k. For example, if a mass weighing 10 lb stretches a spring $\frac{1}{2}$ ft, then $10 \quad k(\frac{1}{2})$ implies $k \quad 20$ lb/ft. Necessarily then, a mass weighing, say, 8 lb stretches the same spring only $\frac{2}{5}$ ft.

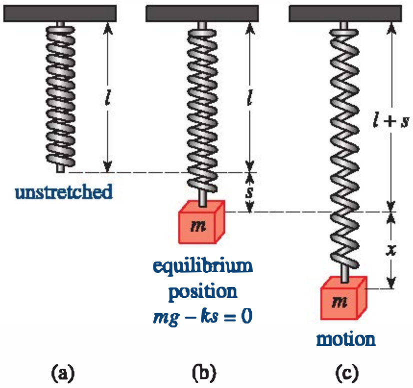


FIGURE 3.8.1 Spring/mass system

Newton's Second Law After a mass m is attached to a spring, it stretches the spring by an amount s and attains a position of equilibrium at which its weight W is balanced by the restoring force ks. Recall that weight is defined by $W \quad mg$, where mass is measured in slugs, kilograms, or grams and $g \quad 32$ ft/s², 9.8 m/s², or 980 cm/s², respectively. As indicated in **FIGURE 3.8.1(b)**, the condition of equilibrium is $mg \quad ks$ or $mg - ks \quad 0$. If the mass is displaced by an amount x from its equilibrium position, the restoring force of the spring is then $k(x + s)$. Assuming that there are no retarding forces acting on the system and assuming that the mass vibrates free of other external forces—**free motion**—we can equate Newton's second law with the net, or resultant, force of the restoring force and the weight:

$$m\frac{d^2x}{dt^2} \quad -k(s + x) + mg \quad -kx + \underbrace{mg - ks}_{\text{zero}} \quad -kx. \tag{1}$$

The negative sign in (1) indicates that the restoring force of the spring acts opposite to the direction of motion. Furthermore, we can adopt the convention that displacements measured *below* the equilibrium position are positive. See **FIGURE 3.8.2**.

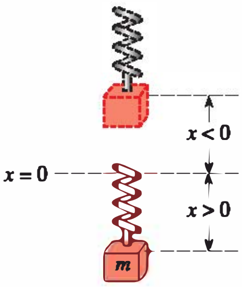


FIGURE 3.8.2 Positive direction is below equilibrium position

DE of Free Undamped Motion

By dividing (1) by the mass m we obtain the second-order differential equation $d^2x/dt^2 + (k/m)x \quad 0$ or

$$\frac{d^2x}{dt^2} + \omega^2 x = 0, \tag{2}$$

where $\omega^2 \quad k/m$. Equation (2) is said to describe **simple harmonic motion** or **free undamped motion**. Two obvious initial conditions associated with (2) are $x(0) \quad x_0$, the amount of initial displacement, and $x'(0) \quad x_1$, the initial velocity of the mass. For example, if $x_0 > 0$, $x_1 < 0$, the mass starts from a point *below* the equilibrium position with an imparted *upward* velocity. When $x_1 \quad 0$ the mass is said to be released from *rest*. For example, if $x_0 < 0$, $x_1 \quad 0$, the mass is released from rest from a point $|x_0|$ units *above* the equilibrium position.

Solution and Equation of Motion

To solve equation (2) we note that the solutions of the auxiliary equation $m^2 + \omega^2 \quad 0$ are the complex numbers $m_1 \quad \omega i$, $m_2 \quad -\omega i$. Thus from (8) of Section 3.3 we find the general solution of (2) to be

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t. \tag{3}$$

The **period of free vibrations** described by (3) is $T \quad 2\pi/\omega$, and the **frequency** is $f \quad 1/T \quad \omega/2\pi$. For example, for $x(t) \quad 2\cos 3t - 4\sin 3t$ the period is $2\pi/3$ and the frequency is $3/2\pi$. The former number means that the graph of $x(t)$ repeats every $2\pi/3$ units; the latter number means that there are three cycles of the graph every 2π units or, equivalently, that the mass undergoes $3/2\pi$ complete vibrations per unit time. In addition, it can be shown that the period $2\pi/\omega$ is the time interval between two successive maxima of $x(t)$. Keep in mind that a maximum of $x(t)$ is a positive displacement corresponding to the mass's attaining a maximum distance *below* the equilibrium position, whereas a minimum of $x(t)$ is a negative displacement corresponding to the mass's attaining a maximum height *above* the equilibrium position. We refer to either case as an **extreme displacement** of the mass. Finally, when the initial conditions are used to determine the constants c_1 and c_2 in (3), we say that the resulting particular solution or response is the **equation of motion**.

EXAMPLE 1 Free Undamped Motion

A mass weighing 2 pounds stretches a spring 6 inches. At $t \quad 0$ the mass is released from a point 8 inches below the equilibrium position with an upward velocity of $\frac{4}{3}$ ft/s. Determine the equation of free motion.

SOLUTION Because we are using the engineering system of units, the measurements given in terms of inches must be converted into feet: 6 in $\quad \frac{1}{2}$ ft; 8 in $\quad \frac{2}{3}$ ft. In addition, we must convert the units of weight given in pounds into units of mass. From $m \quad W/g$ we have $m \quad \frac{2}{32} \quad \frac{1}{16}$ slug. Also, from Hooke's law, $2 \quad k(\frac{1}{2})$ implies that the spring constant is $k \quad 4$ lb/ft. Hence (1) gives

$$\frac{1}{16}\frac{d^2x}{dt^2} = -4x \quad \text{or} \quad \frac{d^2x}{dt^2} + 64x \quad 0.$$

The initial displacement and initial velocity are $x(0) \quad \frac{2}{3}$, $x'(0) \quad -\frac{4}{3}$, where the negative sign in the last condition is a consequence of the fact that the mass is given an initial velocity in the negative, or upward, direction.

Now $\omega^2 \quad 64$ or $\omega \quad 8$, so that the general solution of the differential equation is

$$x(t) \quad c_1 \cos 8t + c_2 \sin 8t. \tag{4}$$

Applying the initial conditions to $x(t)$ and $x'(t)$ gives $c_1 \quad \frac{2}{3}$ and $c_2 \quad -\frac{1}{6}$. Thus the equation of motion is

$$x(t) \quad \frac{2}{3}\cos 8t - \frac{1}{6}\sin 8t. \tag{5}$$

≡

Alternative Form of $x(t)$

When $c_1 \neq 0$ and $c_2 \neq 0$, the actual **amplitude** A of free vibrations is not obvious from inspection of equation (3). For example, although the mass in Example 1 is initially displaced $\frac{2}{3}$ foot beyond the equilibrium position, the amplitude of vibrations is a number larger than $\frac{2}{3}$. Hence it is often convenient to convert an equation of simple harmonic motion of the form given in (3) into the form of a shifted sine function (6) or a shifted cosine function (6′). In both (6) and (6′) the number $A = \sqrt{c_1^2 + c_2^2}$ is the **amplitude of free vibrations**, and ϕ is a **phase angle**. Note carefully that ϕ is defined in a slightly different manner in (7) and (7′).

$$y = A\sin(\omega t + \phi) \tag{6}$$

where

$$\left.\begin{aligned}\sin\phi &= \frac{c_1}{A}\\ \cos\phi &= \frac{c_2}{A}\end{aligned}\right\}\tan\phi = \frac{c_1}{c_2} \tag{7}$$

$$y = A\cos(\omega t - \phi) \tag{6′}$$

where

$$\left.\begin{aligned}\sin\phi &= \frac{c_2}{A}\\ \cos\phi &= \frac{c_1}{A}\end{aligned}\right\}\tan\phi = \frac{c_2}{c_1} \tag{7′}$$

To verify (6), we expand $\sin(\omega t + \phi)$ by the addition formula for the sine function:

$$A\sin(\omega t + \phi) \quad A\sin\omega t\cos\phi + A\cos\omega t\sin\phi \quad (A\sin\phi)\cos\omega t + (A\cos\phi)\sin\omega t. \tag{8}$$

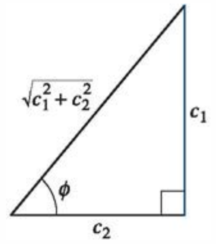


FIGURE 3.8.3 A relationship between $c_1 > 0$, $c_2 > 0$ and phase angle ϕ

It follows from **FIGURE 3.8.3** that if ϕ is defined by

$$\sin\phi \quad \frac{c_1}{\sqrt{c_1^2 + c_2^2}} \quad \frac{c_1}{A}, \quad \cos\phi \quad \frac{c_2}{\sqrt{c_1^2 + c_2^2}} \quad \frac{c_2}{A},$$

then (8) becomes

$$A\frac{c_1}{A}\cos\omega t + A\frac{c_2}{A}\sin\omega t \quad c_1\cos\omega t + c_2\sin\omega t \quad x(t).$$

EXAMPLE 2 Alternative Form of Solution (5)

In view of the foregoing discussion, we can write the solution (5) in the alternative forms (6) and (6′). In both cases the amplitude is $A = \sqrt{(\frac{2}{3})^2 + (-\frac{1}{6})^2} = \sqrt{\frac{17}{36}} \approx 0.69$ ft. But some care should be exercised when computing a phase angle ϕ.

Be careful in the computation of the phase angle ϕ.

(a) With $c_1 \quad \frac{2}{3}$ and $c_2 = -\frac{1}{6}$ we find from (7) that $\tan\phi = -4$. A calculator then gives $\tan^{-1}(-4) = -1.326$ rad. However, this is *not* the phase angle since $\tan^{-1}(-4)$ is located in the *fourth quadrant* and therefore contradicts the fact that $\sin\phi > 0$ and $\cos\phi < 0$ because $c_1 > 0$ and $c_2 < 0$. Hence we must take ϕ to be the *second-quadrant* angle $\phi \quad \pi + (1.326) \quad 1.816$ rad. Thus, (6) gives

$$x(t) \quad \frac{\sqrt{17}}{6}\sin(8t + 1.816). \tag{9}$$

The period of this function is $T \quad 2\pi/8 \quad \pi/4$.

(b) Now with $c_1 \quad \frac{2}{3}$ and $c_2 = -\frac{1}{6}$, we see that (7′) indicates that $\sin\phi < 0$ and $\cos\phi > 0$ and so the angle ϕ lies in the fourth quadrant. Hence from $\tan\phi = -\frac{1}{4}$ we can take $\phi \quad \tan^{-1}(-\frac{1}{4}) = -0.245$ rad. A second alternative form of solution (5) is then

$$x(t) \quad \frac{\sqrt{17}}{6}\cos(8t - (-0.245)) \quad \text{or} \quad x(t) \quad \frac{\sqrt{17}}{6}\cos(8t + 0.245). \tag{9′}$$

≡

Graphical Interpretation

FIGURE 3.8.4(a) illustrates the mass in Example 2 going through approximately two complete cycles of motion. Reading left to right, the first five positions marked with black dots in the figure correspond, respectively,

- to the initial position ($x \quad \frac{2}{3}$) of the mass below the equilibrium position,
- to the mass passing through the equilibrium position ($x \quad 0$) for the first time heading upward,
- to the mass at its extreme displacement ($x = -\frac{\sqrt{17}}{6}$) above the equilibrium position,
- to the mass passing through the equilibrium position ($x \quad 0$) for the second time heading downward, and
- to the mass at its extreme displacement ($x \quad \frac{\sqrt{17}}{6}$) below the equilibrium position.

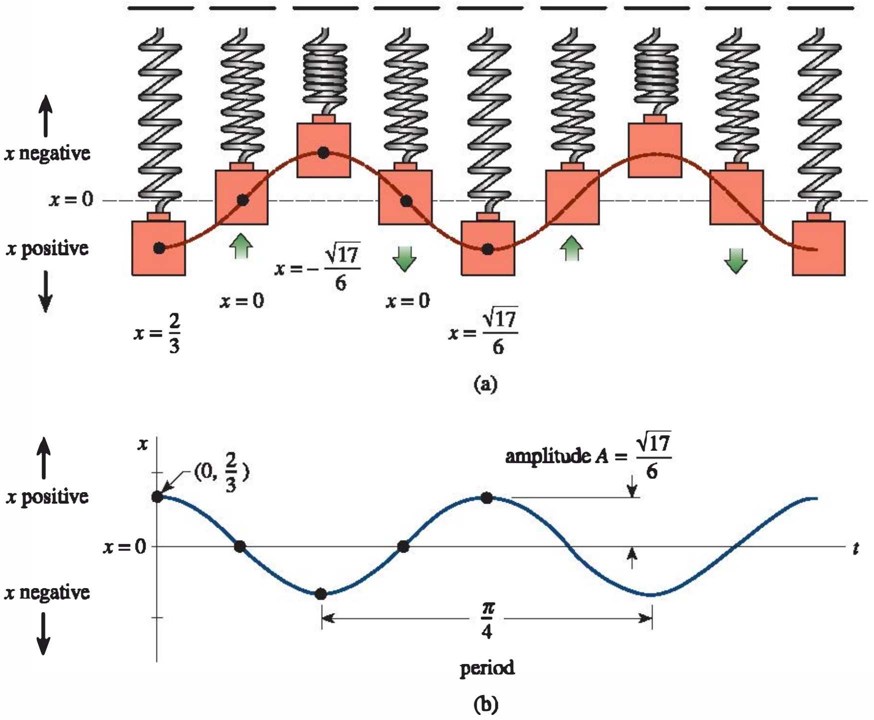


FIGURE 3.8.4 Simple harmonic motion

The dots on the graph of (9) given in Figure 3.8.4(b) also agree with the five positions just given. Note, however, that in Figure 3.8.4(b) the positive direction in the tx-plane is the usual upward direction and so is opposite to the positive direction indicated in Figure 3.8.4(a). Hence the blue graph representing the motion of the mass in Figure 3.8.4(b) is the mirror image through the t-axis of the red dashed curve in Figure 3.8.4(a).

Form (6) is very useful, since it is easy to find values of time for which the graph of $x(t)$ crosses the positive t-axis (the line $x \quad 0$). We observe that $\sin(\omega t + \phi) \quad 0$ when $\omega t + \phi \quad n\pi$, where n is a nonnegative integer.

Systems with Variable Spring Constants In the model discussed above, we assumed an ideal world, a world in which the physical characteristics of the spring do not change over time. In the nonideal world, however, it seems reasonable to expect that when a spring/mass system is in motion for a long period the spring would weaken; in other words, the "spring constant" would vary, or, more specifically, decay with time. In one model for the **aging spring**, the spring constant k in (1), is replaced by the decreasing function $K(t) \quad ke^{-\alpha t}$, $k > 0$, $\alpha > 0$. The linear differential equation $mx'' + ke^{-\alpha t}x \quad 0$ cannot be solved by the methods considered in this chapter. Nevertheless, we can obtain two linearly independent solutions using the methods in Chapter 5. See Problem 15 in Exercises 3.8, Example 4 in Section 5.3, and Problems 33 and 43 in Exercises 5.3.

When a spring/mass system is subjected to an environment in which the temperature is rapidly decreasing, it might make sense to replace the constant k with $K(t) \quad kt$, $k > 0$, a function that increases with time. The resulting model, $mx'' + ktx \quad 0$, is a form of **Airy's differential equation**. Like the equation for an aging spring, Airy's equation can be solved by the methods of Chapter 5. See Problem 16 in Exercises 3.8, Example 2 in Section 5.1, and Problems 34, 35, and 44 in Exercises 5.3.

3.8.2 Spring/Mass Systems: Free Damped Motion

The concept of free harmonic motion is somewhat unrealistic, since the motion described by equation (1) assumes that there are no retarding forces acting on the moving mass. Unless the mass is suspended in a perfect vacuum, there will be at least a resisting force due to the surrounding medium. As **FIGURE 3.8.5** shows, the mass could be suspended in a viscous medium or connected to a dashpot damping device.

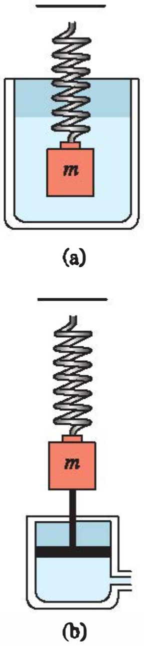


FIGURE 3.8.5 Damping devices

□ DE of Free Damped Motion In the study of mechanics, damping forces acting on a body are considered to be proportional to a power of the instantaneous velocity. In particular, we shall assume throughout the subsequent discussion that this force is given by a constant multiple of dx/dt. When no other external forces are impressed on the system, it follows from Newton's second law that

$$m\frac{d^2x}{dt^2} = -kx - \beta\frac{dx}{dt}, \tag{10}$$

where β is a positive *damping constant* and the negative sign is a consequence of the fact that the damping force acts in a direction opposite to the motion.

Dividing (10) by the mass m, we find the differential equation of **free damped motion** is $d^2x/dt^2 + (\beta/m)dx/dt + (k/m)x \quad 0$ or

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \omega^2 x = 0, \tag{11}$$

where $$2\lambda \quad \frac{\beta}{m}, \quad \omega^2 \quad \frac{k}{m}. \tag{12}$$

The symbol 2λ is used only for algebraic convenience, since the auxiliary equation is $m^2 + 2\lambda m + \omega^2 \quad 0$ and the corresponding roots are then

$$m_1 = -\lambda + \sqrt{\lambda^2 - \omega^2}, \quad m_2 = -\lambda - \sqrt{\lambda^2 - \omega^2}.$$

We can now distinguish three possible cases depending on the algebraic sign of $\lambda^2 - \omega^2$. Since each solution contains the *damping factor* $e^{-\lambda t}$, $\lambda > 0$, the displacements of the mass become negligible over a long period of time.

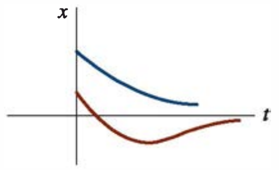


FIGURE 3.8.6 Motion of an overdamped system

***Case I*: $\lambda^2 - \omega^2 > 0$** In this situation the system is said to be **overdamped** because the damping coefficient β is large when compared to the spring constant k. The corresponding solution of (11) is $x(t) \quad c_1e^{m_1t} + c_2e^{m_2t}$ or

$$x(t) = e^{-\lambda t}\left(c_1e^{\sqrt{\lambda^2-\omega^2}t} + c_2e^{-\sqrt{\lambda^2-\omega^2}t}\right). \tag{13}$$

Equation 13 represents a smooth and nonoscillatory motion. **FIGURE 3.8.6** shows two possible graphs of $x(t)$.

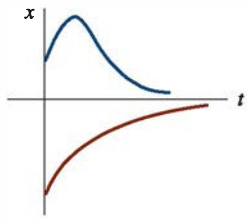


FIGURE 3.8.7 Motion of an critically damped system

***Case II*: $\lambda^2 - \omega^2 = 0$** The system is said to be **critically damped** because any slight decrease in the damping force would result in oscillatory motion. The general solution of (11) is $x(t) \quad c_1e^{m_1t} + c_2te^{m_1t}$ or

$$x(t) = e^{-\lambda t}(c_1 + c_2t). \tag{14}$$

Some graphs of typical motion are given in **FIGURE 3.8.7**. Notice that the motion is quite similar to that of an overdamped system. It is also apparent from (14) that the mass can pass through the equilibrium position at most one time.

***Case III*: $\lambda^2 - \omega^2 < 0$** In this case the system is said to be **underdamped** because the damping coefficient is small compared to the spring constant. The roots m_1 and m_2 are now complex:

$$m_1 = -\lambda + \sqrt{\omega^2 - \lambda^2}i, \quad m_2 = -\lambda - \sqrt{\omega^2 - \lambda^2}i.$$

Thus the general solution of equation (11) is

$$x(t) = e^{-\lambda t}\left(c_1\cos\sqrt{\omega^2 - \lambda^2}t + c_2\sin\sqrt{\omega^2 - \lambda^2}t\right). \tag{15}$$

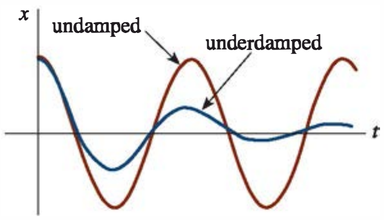


FIGURE 3.8.8 Motion of an underdamped system

As indicated in **FIGURE 3.8.8**, the motion described by (15) is oscillatory, but because of the coefficient $e^{-\lambda t}$, the amplitudes of vibration $\to 0$ as $t \to \infty$.

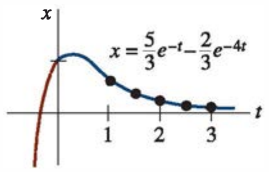
EXAMPLE 3 Overdamped Motion

It is readily verified that the solution of the initial-value problem

$$\frac{d^2x}{dt^2} + 5\frac{dx}{dt} + 4x \quad 0, \quad x(0) \quad 1, \quad x'(0) \quad 1$$

is

$$x(t) \quad \frac{5}{3}e^{-t} - \frac{2}{3}e^{-4t}. \tag{16}$$



(a)

t	$x(t)$
1	0.601
1.5	0.370
2	0.225
2.5	0.137
3	0.083

(b)

FIGURE 3.8.9 Overdamped system in Example 3

The problem can be interpreted as representing the overdamped motion of a mass on a spring. The mass starts from a position 1 unit *below* the equilibrium position with a *downward* velocity of 1 ft/s.

To graph $x(t)$ we find the value of t for which the function has an extremum—that is, the value of time for which the first derivative (velocity) is zero. Differentiating (16) gives $x'(t) \quad -\frac{5}{3}e^{-t} + \frac{8}{3}e^{-4t}$ so that $x'(t) \quad 0$ implies $e^{3t} \quad \frac{8}{5}$ or $t \quad \frac{1}{3}\ln\frac{8}{5} \quad 0.157$. It follows from the first derivative test, as well as our intuition, that $x(0.157) \quad 1.069$ ft is actually a maximum. In other words, the mass attains an extreme displacement of 1.069 feet below the equilibrium position.

We should also check to see whether the graph crosses the t-axis; that is, whether the mass passes through the equilibrium position. This cannot happen in this instance since the equation $x(t) \quad 0$, or $e^{3t} \quad \frac{2}{5}$, has the physically irrelevant solution $t \quad \frac{1}{3}\ln\frac{2}{5} \quad -0.305$.

The graph of $x(t)$, along with some other pertinent data, is given in **FIGURE 3.8.9**. ≡

EXAMPLE 4 Critically Damped Motion

An 8-pound weight stretches a spring 2 feet. Assuming that a damping force numerically equal to two times the instantaneous velocity acts on the system, determine the equation of motion if the weight is released from the equilibrium position with an upward velocity of 3 ft/s.

SOLUTION From Hooke's law we see that $8 \quad k(2)$ gives $k \quad 4$ lb/ft and that $W \quad mg$ gives $m \quad \frac{8}{32} \quad \frac{1}{4}$ slug. The differential equation of motion is then

$$\frac{1}{4}\frac{d^2x}{dt^2} = -4x - 2\frac{dx}{dt} \quad \text{or} \quad \frac{d^2x}{dt^2} + 8\frac{dx}{dt} + 16x \quad 0. \tag{17}$$

The auxiliary equation for (17) is $m^2 + 8m + 16 \quad (m + 4)^2 \quad 0$ so that $m_1 \quad m_2 \quad -4$. Hence the system is critically damped and

$$x(t) \quad c_1e^{-4t} + c_2te^{-4t}. \tag{18}$$

Applying the initial conditions $x(0) \quad 0$ and $x'(0) \quad -3$, we find, in turn, that $c_1 \quad 0$ and $c_2 \quad -3$. Thus the equation of motion is

$$x(t) \quad -3te^{-4t}. \tag{19}$$

To graph $x(t)$ we proceed as in Example 3. From $x'(t) \quad -3e^{-4t}(1 - 4t)$ we see that $x'(t) \quad 0$ when $t \quad \frac{1}{4}$. The corresponding extreme displacement is $x(\frac{1}{4}) \quad -3(\frac{1}{4})e^{-1} \quad -0.276$ ft. As shown in **FIGURE 3.8.10**, we interpret this value to mean that the weight reaches a maximum height of 0.276 foot above the equilibrium position. ≡

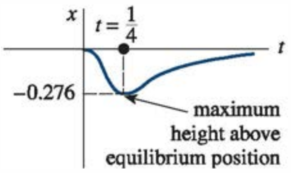


FIGURE 3.8.10 Critically damped system in Example 4

EXAMPLE 5 Underdamped Motion

A 16-pound weight is attached to a 5-foot-long spring. At equilibrium the spring measures 8.2 feet. If the weight is pushed up and released from rest at a point 2 feet above the equilibrium position, find the displacements $x(t)$ if it is further known that the surrounding medium offers a resistance numerically equal to the instantaneous velocity.

SOLUTION The elongation of the spring after the weight is attached is $8.2 - 5 \quad 3.2$ ft, so it follows from Hooke's law that $16 \quad k(3.2)$ or $k \quad 5$ lb/ft. In addition, $m \quad \frac{16}{32} \quad \frac{1}{2}$ slug so that the differential equation is given by

$$\frac{1}{2}\frac{d^2x}{dt^2} = -5x - \frac{dx}{dt} \quad \text{or} \quad \frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 10x \quad 0. \tag{20}$$

Proceeding, we find that the roots of $m^2 + 2m + 10 \quad 0$ are $m_1 \quad -1 + 3i$ and $m_2 \quad -1 - 3i$, which then implies the system is underdamped and

$$x(t) \quad e^{-t}(c_1 \cos 3t + c_2 \sin 3t). \tag{21}$$

Finally, the initial conditions $x(0) \quad -2$ and $x'(0) \quad 0$ yield $c_1 \quad -2$ and $c_2 \quad -\frac{2}{3}$, so the equation of motion is

$$x(t) \quad e^{-t}\left(-2\cos 3t - \frac{2}{3}\sin 3t\right). \tag{22}$$ ≡

Alternative Form of $x(t)$ In a manner identical to the procedure used on page 152, we can write any solution

$$x(t) \quad e^{-\lambda t}(c_1 \cos\sqrt{\omega^2 - \lambda^2}t + c_2 \sin\sqrt{\omega^2 - \lambda^2}t)$$

in the alternative form

$$x(t) = Ae^{-\lambda t}\sin(\sqrt{\omega^2 - \lambda^2}t + \phi), \tag{23}$$

where $A \quad \sqrt{c_1^2 + c_2^2}$ and the phase angle ϕ is determined from the equations

$$\sin\phi \quad \frac{c_1}{A}, \quad \cos\phi \quad \frac{c_2}{A}, \quad \tan\phi \quad \frac{c_1}{c_2}.$$

The coefficient $Ae^{-\lambda t}$ is sometimes called the **damped amplitude** of vibrations. Because (23) is not a periodic function, the number $2\pi/\sqrt{\omega^2 - \lambda^2}$ is called the **quasi period** and $\sqrt{\omega^2 - \lambda^2}/2\pi$ is the **quasi frequency**. The quasi period is the time interval between two successive maxima of $x(t)$. You should verify, for the equation of motion in Example 5, that $A \quad 2\sqrt{10}/3$ and $\phi \quad 4.391$. Therefore an equivalent form of (22) is

$$x(t) \quad \frac{2\sqrt{10}}{3}e^{-t}\sin(3t + 4.391).$$

3.8.3 Spring/Mass Systems: Driven Motion

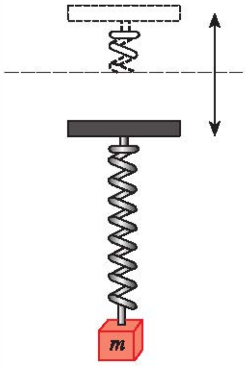


FIGURE 3.8.11 Oscillatory vertical motion of the support

DE of Driven Motion with Damping Suppose we now take into consideration an external force $f(t)$ acting on a vibrating mass on a spring. For example, $f(t)$ could represent a driving force causing an oscillatory vertical motion of the support of the spring. See **FIGURE 3.8.11**. The inclusion of $f(t)$ in the formulation of Newton's second law gives the differential equation of **driven** or **forced motion**:

$$m\frac{d^2x}{dt^2} = -kx - \beta\frac{dx}{dt} + f(t). \tag{24}$$

Dividing (24) by m gives

$$\frac{d^2x}{dt^2} + 2\lambda\frac{dx}{dt} + \omega^2 x = F(t), \tag{25}$$

where $F(t) \quad f(t)/m$ and, as in the preceding section, $2\lambda \quad \beta/m$, $\omega^2 \quad k/m$. To solve the latter nonhomogeneous equation we can use either the method of undetermined coefficients or variation of parameters.

EXAMPLE 6 Interpretation of an Initial-Value Problem

Interpret and solve the initial-value problem

$$\frac{1}{5}\frac{d^2x}{dt^2} + 1.2\frac{dx}{dt} + 2x \quad 5\cos 4t, \quad x(0) \quad \frac{1}{2}, \quad x'(0) \quad 0. \tag{26}$$

SOLUTION We can interpret the problem to represent a vibrational system consisting of a mass ($m \quad \frac{1}{5}$ slug or kilogram) attached to a spring ($k \quad 2$ lb/ft or N/m). The mass is released from rest $\frac{1}{2}$ unit (foot or meter) below the equilibrium position. The motion is damped ($\beta \quad 1.2$) and is being driven by an external periodic ($T \quad \pi/2$ s) force beginning at $t \quad 0$. Intuitively we would expect that even with damping, the system would remain in motion until such time as the forcing function was "turned off," in which case the amplitudes would diminish. However, as the problem is given, $f(t) \quad 5\cos 4t$ will remain "on" forever.

We first multiply the differential equation in (26) by 5 and solve

$$\frac{dx^2}{dt^2} + 6\frac{dx}{dt} + 10x \quad 0$$

by the usual methods. Since $m_1 \quad -3 + i$, $m_2 \quad -3 - i$, it follows that

$$x_c(t) \quad e^{-3t}(c_1 \cos t + c_2 \sin t).$$

Using the method of undetermined coefficients, we assume a particular solution of the form $x_p(t) \quad A\cos 4t + B\sin 4t$. Differentiating $x_p(t)$ and substituting into the DE gives

$$x_p'' + 6x_p' + 10x_p \quad (-6A + 24B)\cos 4t + (-24A - 6B)\sin 4t \quad 25\cos 4t.$$

The resulting system of equations

$$-6A + 24B \quad 25, \qquad -24A - 6B \quad 0$$

yields $A \quad -\frac{25}{102}$ and $B \quad \frac{50}{51}$. It follows that

$$x(t) \quad e^{-3t}(c_1 \cos t + c_2 \sin t) - \frac{25}{102}\cos 4t + \frac{50}{51}\sin 4t. \tag{27}$$

When we set $t \quad 0$ in the above equation, we obtain $c_1 \quad \frac{38}{51}$. By differentiating the expression and then setting $t \quad 0$, we also find that $c_2 \quad -\frac{86}{51}$. Therefore the equation of motion is

$$x(t) \quad e^{-3t}\left(\frac{38}{51}\cos t - \frac{86}{51}\sin t\right) - \frac{25}{102}\cos 4t + \frac{50}{51}\sin 4t. \tag{28}$$ ≡

Transient and Steady-State Terms When F is a periodic function, such as $F(t) \quad F_0 \sin \gamma t$ or $F(t) \quad F_0 \cos \gamma t$, the general solution of (25) for $\lambda > 0$ is the sum of a nonperiodic function $x_c(t)$ and a periodic function $x_p(t)$. Moreover, $x_c(t)$ dies off as time increases; that is, $\lim_{t\to\infty} x_c(t) \quad 0$. Thus for a long period of time, the displacements of the mass are closely approximated by the particular solution $x_p(t)$. The complementary function $x_c(t)$ is said to be a **transient term** or **transient solution**, and the function $x_p(t)$, the part of the solution that remains after an interval of time, is called a **steady-state term** or **steady-state solution**. Note therefore that the effect of the initial conditions on a spring/mass system driven by F is transient. In the particular solution (28), $e^{-3t}(\frac{38}{51}\cos t - \frac{86}{51}\sin t)$ is a transient term and $x_p(t) \quad -\frac{25}{102}\cos 4t + \frac{50}{51}\sin 4t$ is a steady-state term. The graphs of these two terms and the solution (28) are given in FIGURES 3.8.12(a) and 3.8.12(b), respectively.

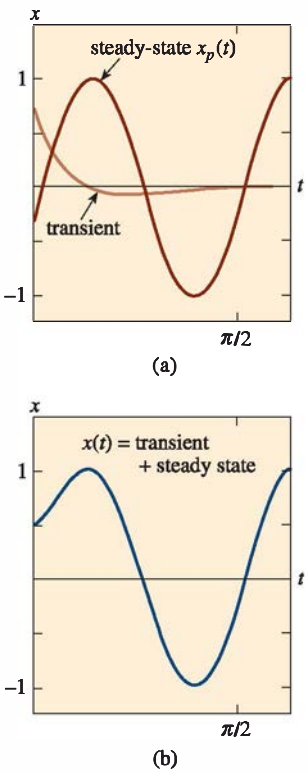


FIGURE 3.8.12 Graph of solution (28) in Example 6

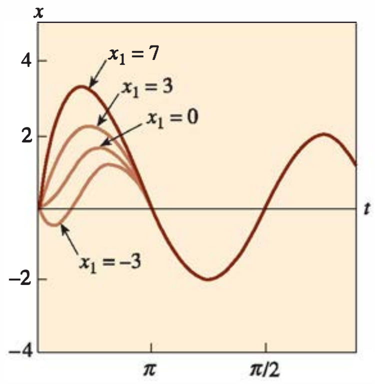


FIGURE 3.8.13 Graphs of solution in Example 7 for various values of x_1

EXAMPLE 7 Transient/Steady-State Solutions

The solution of the initial-value problem

$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 2x \quad 4\cos t + 2\sin t, \quad x(0) \quad 0, \quad x'(0) \quad x_1,$$

where x_1 is constant, is given by

$$x(t) \quad \underbrace{(x_1 - 2)e^{-t}\sin t}_{\text{transient}} + \underbrace{2\sin t}_{\text{steady state}}.$$

Solution curves for selected values of the initial velocity x_1 are shown in **FIGURE 3.8.13**. The graphs show that the influence of the transient term is negligible for about $t > 3\pi/2$. ≡

DE of Driven Motion Without Damping

With a periodic impressed force and no damping force, there is no transient term in the solution of a problem. Also, we shall see that a periodic impressed force with a frequency near or the same as the frequency of free undamped vibrations can cause a severe problem in any oscillatory mechanical system.

EXAMPLE 8 Undamped Forced Motion

Solve the initial-value problem

$$\frac{d^2x}{dt^2} + \omega^2 x \quad F_0 \sin \gamma t, \quad x(0) \quad 0, \quad x'(0) \quad 0, \tag{29}$$

where F_0 is a constant and $\gamma \neq \omega$.

SOLUTION The complementary function is $x_c(t) \quad c_1 \cos \omega t + c_2 \sin \omega t$. To obtain a particular solution we assume $x_p(t) \quad A\cos\gamma t + B\sin\gamma t$ so that

$$x_p'' + \omega^2 x_p \quad A(\omega^2 - \gamma^2)\cos\gamma t + B(\omega^2 - \gamma^2)\sin\gamma t \quad F_0 \sin\gamma t.$$

Equating coefficients immediately gives $A \quad 0$ and $B \quad F_0/(\omega^2 - \gamma^2)$. Therefore

$$x_p(t) \quad \frac{F_0}{\omega^2 - \gamma^2}\sin\gamma t.$$

Applying the given initial conditions to the general solution

$$x(t) \quad c_1\cos\omega t + c_2 \sin\omega t + \frac{F_0}{\omega^2 - \gamma^2}\sin\gamma t$$

yields $c_1 \quad 0$ and $c_2 \quad -\gamma F_0/\omega(\omega^2 - \gamma^2)$. Thus the solution is

$$x(t) \quad \frac{F_0}{\omega(\omega^2 - \gamma^2)}(-\gamma \sin\omega t + \omega \sin\gamma t), \qquad \gamma \neq \omega. \tag{30}$$ ≡

Pure Resonance

Although equation (30) is not defined for $\gamma \quad \omega$, it is interesting to observe that its limiting value as $\gamma \to \omega$ can be obtained by applying L'Hôpital's rule. This limiting process is analogous to "tuning in" the frequency of the driving force $(\gamma/2\pi)$ to the frequency of free vibrations $(\omega/2\pi)$. Intuitively we expect that over a length of time we should be able to substantially increase the amplitudes of vibration. For $\gamma = \omega$ we define the solution to be

$$x(t) = \lim_{\gamma \to \omega} F_0 \frac{-\gamma \sin \omega t + \omega \sin \gamma t}{\omega(\omega^2 - \gamma^2)} = F_0 \lim_{\gamma \to \omega} \frac{\dfrac{d}{d\gamma}(-\gamma \sin \omega t + \omega \sin \gamma t)}{\dfrac{d}{d\gamma}(\omega^3 - \omega\gamma^2)}$$

$$= F_0 \lim_{\gamma \to \omega} \frac{-\sin \omega t + \omega t \cos \gamma t}{-2\omega\gamma}$$

$$= F_0 \frac{-\sin \omega t + \omega t \cos \omega t}{-2\omega^2}$$

$$= \frac{F_0}{2\omega^2} \sin \omega t - \frac{F_0}{2\omega} t \cos \omega t. \tag{31}$$

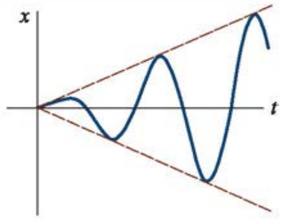


FIGURE 3.8.14 Graph of solution in (31) illustrating pure resonance

As suspected, when $t \to \infty$ the displacements become large; in fact, $|x(t_n)| \to \infty$ when $t_n = n\pi/\omega$, $n = 1, 2, \ldots$. The phenomenon we have just described is known as **pure resonance**. The graph given in **FIGURE 3.8.14** shows typical motion in this case.

In conclusion, it should be noted that there is no actual need to use a limiting process on (30) to obtain the solution for $\gamma = \omega$. Alternatively, equation (31) follows by solving the initial-value problem

$$\frac{d^2x}{dt^2} + \omega^2 x = F_0 \sin \omega t, \quad x(0) = 0, \quad x'(0) = 0$$

directly by conventional methods.

If the displacements of a spring/mass system were actually described by a function such as (31), the system would necessarily fail. Large oscillations of the mass would eventually force the spring beyond its elastic limit. One might argue too that the resonating model presented in Figure 3.8.14 is completely unrealistic, because it ignores the retarding effects of ever-present damping forces. Although it is true that pure resonance cannot occur when the smallest amount of damping is taken into consideration, large and equally destructive amplitudes of vibration (although bounded as $t \to \infty$) can occur. See Problem 43 in Exercises 3.8.

3.8.4 Series Circuit Analogue

□ ***LRC*-Series Circuits** As mentioned in the introduction to this chapter, many different physical systems can be described by a linear second-order differential equation similar to the differential equation of forced motion with damping:

$$m\frac{d^2x}{dt^2} + \beta\frac{dx}{dt} + kx = f(t). \tag{32}$$

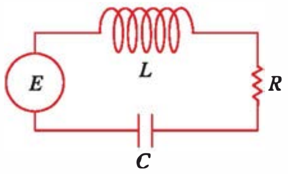


FIGURE 3.8.15 *LRC*-series circuit

If $i(t)$ denotes current in the ***LRC*-series electrical circuit** shown in **FIGURE 3.8.15**, then the voltage drops across the inductor, resistor, and capacitor are as shown in Figure 1.3.5. By Kirchhoff's second law, the sum of these voltages equals the voltage $E(t)$ impressed on the circuit; that is,

$$L\frac{di}{dt} + Ri + \frac{1}{C}q = E(t). \tag{33}$$

But the charge $q(t)$ on the capacitor is related to the current $i(t)$ by $i = dq/dt$, and so (33) becomes the linear second-order differential equation

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t). \tag{34}$$

The nomenclature used in the analysis of circuits is similar to that used to describe spring/mass systems.

If $E(t) = 0$, the **electrical vibrations** of the circuit are said to be **free**. Since the auxiliary equation for (34) is $Lm^2 + Rm + 1/C = 0$, there will be three forms of the solution with $R \neq 0$, depending on the value of the discriminant $R^2 - 4L/C$. We say that the circuit is

	overdamped if	$R^2 - 4L/C > 0$,
	critically damped if	$R^2 - 4L/C = 0$,
and	**underdamped** if	$R^2 - 4L/C < 0$.

In each of these three cases the general solution of (34) contains the factor $e^{-Rt/2L}$, and so $q(t) \to 0$ as $t \to \infty$. In the underdamped case when $q(0) = q_0$, the charge on the capacitor oscillates as it decays; in other words, the capacitor is charging and discharging as $t \to \infty$. When $E(t) = 0$ and $R = 0$, the circuit is said to be undamped and the electrical vibrations do not approach zero as t increases without bound; the response of the circuit is **simple harmonic**.

EXAMPLE 9 Underdamped Series Circuit

Find the charge $q(t)$ on the capacitor in an LRC-series circuit when $L = 0.25$ henry (h), $R = 10$ ohms (Ω), $C = 0.001$ farad (f), $E(t) = 0$ volts (V), $q(0) = q_0$ coulombs (C), and $i(0) = 0$ amperes (A).

SOLUTION Since $1/C = 1000$, equation (34) becomes

$$\frac{1}{4}q'' + 10q' + 1000q = 0 \quad \text{or} \quad q'' + 40q' + 4000q = 0.$$

Solving this homogeneous equation in the usual manner, we find that the circuit is underdamped and $q(t) = e^{-20t}(c_1 \cos 60t + c_2 \sin 60t)$. Applying the initial conditions, we find $c_1 = q_0$ and $c_2 = q_0/3$. Thus $q(t) = q_0 e^{-20t}(\cos 60t + \frac{1}{3}\sin 60t)$. Using (23), we can write the foregoing solution as

$$q(t) = \frac{q_0\sqrt{10}}{3}e^{-20t}\sin(60t + 1.249). \qquad \equiv$$

When there is an impressed voltage $E(t)$ on the circuit, the electrical vibrations are said to be **forced**. In the case when $R \neq 0$, the complementary function $q_c(t)$ of (34) is called a **transient solution**. If $E(t)$ is periodic or a constant, then the particular solution $q_p(t)$ of (34) is a **steady-state solution**.

EXAMPLE 10 Steady-State Current

Find the steady-state solution $q_p(t)$ and the **steady-state current** in an LRC-series circuit when the impressed voltage is $E(t) = E_0 \sin \gamma t$.

SOLUTION The steady-state solution $q_p(t)$ is a particular solution of the differential equation

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E_0 \sin \gamma t.$$

Using the method of undetermined coefficients, we assume a particular solution of the form $q_p(t) = A \sin \gamma t + B \cos \gamma t$. Substituting this expression into the differential equation, simplifying, and equating coefficients gives

$$A = \frac{E_0\left(L\gamma - \dfrac{1}{C\gamma}\right)}{-\gamma\left(L^2\gamma^2 - \dfrac{2L}{C} + \dfrac{1}{C^2\gamma^2} + R^2\right)}, \quad B = \frac{E_0 R}{-\gamma\left(L^2\gamma^2 - \dfrac{2L}{C} + \dfrac{1}{C^2\gamma^2} + R^2\right)}.$$

It is convenient to express A and B in terms of some new symbols.

$$\text{If } X = L\gamma - \frac{1}{C\gamma}, \quad \text{then} \quad X^2 = L^2\gamma^2 - \frac{2L}{C} + \frac{1}{C^2\gamma^2}.$$

$$\text{If } Z = \sqrt{X^2 + R^2}, \quad \text{then} \quad Z^2 = L^2\gamma^2 - \frac{2L}{C} + \frac{1}{C^2\gamma^2} + R^2.$$

Therefore $A = E_0X/(-\gamma Z^2)$ and $B = E_0R/(-\gamma Z^2)$, so the steady-state charge is

$$q_p(t) = -\frac{E_0X}{\gamma Z^2}\sin\gamma t - \frac{E_0R}{\gamma Z^2}\cos\gamma t.$$

Now the steady-state current is given by $i_p(t) = q_p'(t)$:

$$i_p(t) = \frac{E_0}{Z}\left(\frac{R}{Z}\sin\gamma t - \frac{X}{Z}\cos\gamma t\right). \qquad (35) \equiv$$

The quantities $X = L\gamma - 1/(C\gamma)$ and $Z = \sqrt{X^2 + R^2}$ defined in Example 10 are called, respectively, the **reactance** and **impedance** of the circuit. Both the reactance and the impedance are measured in ohms.

3.8 Exercises

Answers to selected odd-numbered problems begin on page ANS-6.

3.8.1 Spring/Mass Systems: Free Undamped Motion

1. A mass weighing 4 pounds is attached to a spring whose spring constant is 16 lb/ft. What is the period of simple harmonic motion?
2. A 20-kilogram mass is attached to a spring. If the frequency of simple harmonic motion is $2/\pi$ cycles/s, what is the spring constant k? What is the frequency of simple harmonic motion if the original mass is replaced with an 80-kilogram mass?
3. A mass weighing 24 pounds, attached to the end of a spring, stretches it 4 inches. Initially, the mass is released from rest from a point 3 inches above the equilibrium position. Find the equation of motion.
4. Determine the equation of motion if the mass in Problem 3 is initially released from the equilibrium position with an initial downward velocity of 2 ft/s.
5. A mass weighing 20 pounds stretches a spring 6 inches. The mass is initially released from rest from a point 6 inches below the equilibrium position.
   (a) Find the position of the mass at the times $t = \pi/12, \pi/8, \pi/6, \pi/4$, and $9\pi/32$ s.
   (b) What is the velocity of the mass when $t = 3\pi/16$ s? In which direction is the mass heading at this instant?
   (c) At what times does the mass pass through the equilibrium position?
6. A force of 400 newtons stretches a spring 2 meters. A mass of 50 kilograms is attached to the end of the spring and is initially released from the equilibrium position with an upward velocity of 10 m/s. Find the equation of motion.
7. Another spring whose constant is 20 N/m is suspended from the same rigid support but parallel to the spring/mass system in Problem 6. A mass of 20 kilograms is attached to the second spring, and both masses are initially released from the equilibrium position with an upward velocity of 10 m/s.
   (a) Which mass exhibits the greater amplitude of motion?
   (b) Which mass is moving faster at $t = \pi/4$ s? At $\pi/2$ s?
   (c) At what times are the two masses in the same position? Where are the masses at these times? In which directions are they moving?
8. A mass weighing 32 pounds stretches a spring 2 feet.
   (a) Determine the amplitude and period of motion if the mass is initially released from a point 1 foot above the equilibrium position with a upward velocity of 2 ft/s.
   (b) How many complete cycles will the mass have made at the end of 4π seconds?
9. A mass weighing 8 pounds is attached to a spring. When set in motion, the spring/mass system exhibits simple harmonic motion.
   (a) Determine the equation of motion if the spring constant is 1 lb/ft and the mass is initially released from a point 6 inches below the equilibrium position with a downward velocity of $\frac{3}{2}$ ft/s.
   (b) Express the equation of motion in the form of a shifted sine function given in (6).
   (c) Express the equation of motion in the form of a shifted cosine function given in (6′).
10. A mass weighing 10 pounds stretches a spring $\frac{1}{4}$ foot. This mass is removed and replaced with a mass of 1.6 slugs, which is initially released from a point $\frac{1}{3}$ foot above the equilibrium position with a downward velocity of $\frac{5}{4}$ ft/s.
    (a) Express the equation of motion in the form of a shifted sine function given in (6).
    (b) Express the equation of motion in the form of a shifted cosine function given in (6′).
    (c) Use one of the solutions obtained in parts (a) and (b) to determine the times the mass attains a displacement below the equilibrium position numerically equal to $\frac{1}{2}$ the amplitude of motion.